

Case Study: New Citizen Science Project to Monitor Malaysia Plant Biodiversity

Malaysia represents one of the world's notable biodiversity hotspots, with a large number of remarkable species and ecosystems, but the region is also one of the most botanically threatened. Population growth and the resulting need for agricultural production contribute to increasing pressure on natural resources and ecosystems. These new and greater pressures, exerted by the results of human activities, require us to rethink the way we manage and conserve biodiversity. In recent years, the emergence of initiatives based on citizen science approaches to biodiversity monitoring has profoundly transformed the way we oversee biodiversity.

Citizen science, an alliance between professional scientists and the general public in scientific research, holds immense potential to augment both the "amount of data" and the "number of contributors" in tropical regions. As part of this project, an alpha version of the citizen science platform is proposed, empowering users with web interfaces to identify Endangered, Rare and Threatened (ERT) plants through an automatic visual identification engine. In addition, the web application will integrate a function akin to a plant encyclopedia, enabling users to explore encountered plant species.

Public and taxonomists' comments and opinions will be diligently collected to further refine the plant identification system.

In 2023, several student teams developed this website to help scientists collect plant data.

Your task now is to redesign the website with additional features to better support the scientists in collecting the plant data effectively.

In the previous project, the teams completed the following requirements:

1. Index page – home page
2. Explorer pages (You may refer to PlantsData document)
3. Identify page
4. Contributor page
5. Enquiry page

The table below was the previous project's requirements.

Feature	Condition
Identify (Assignment 1 & Assignment 2 – enhancement)	The web interface must be able to identify the plant type from the uploaded photo. The output will display the scientific plant name and the common name that people can easily recognize. It will also include how to care for the plant and the right environment for it, including the main location where the plant can be found.
Explorer (Assignment 1)	These pages will display plant information based on 3 classification areas: • Families • Genera • Species Each of the areas is linked to each other, for example: From Families, users can also check on the genera and species based on the input search for Families. It is the same as if the user's search on Genera, and also to check the information on Families and Species based on the Genera search. Each page has the following information: • Photo(s). • Information about the plant. • Origin location of the plant. • Classification areas: Families/Genera/Species
Contributor (Assignment 1)	List of photos uploaded by the users. Under this section, the user can view it like a card display with the following information: • Name of the contributors • Name of the plant • Photo(s) of the plant • Comments (from users)

Previous web interface requirements

The following features were designed in the previous web interface project.

User Module

1. Identify (Assignment 1 and Assignment 2-Enhancement)
 - Allowing the user to upload photo to identify the scientific name, common name, care tips and location of the plant.
2. Explorer (Assignment 1)
 - Exploring on the plant search based on the following:
 - Family (e.g., Dipterocarpaceae)
 - Genera (e.g., Vatica)
 - Species (e.g., The right species)
3. Contributor (Assignment 1)
 - To list/show the photos that were uploaded for further information:
 - Name of the contributors

- Name of the plant
 - Photos of the plant
 - comments
4. Enquiry form (Assignment 1)
- Allowing the user to make enquiry on the plant based on:
 - Family
 - Genera
 - Species

New/Additional Features

Assignment 1

1. Create an Explorer page that links to 3 areas: Families, Genera, and Species.
 - Create 2 links to “Identify Page” and “Contributor Page”.
2. Create the “Identify” search button inside the Identify page.
3. Contributor page
 - Create a link to the “Contribution form” for registered user who wants to contribute to the website.
4. Create a contribution form.

Pages	Description	Complete in Assignment 1 / 2
Main page	A page: • To introduce the project	Assignment 1 (UI)
Explorer	Create a drop-down menu for the 3 areas: • Families • Genera • Species Besides the drop-down menu, create a page that links to the 3 areas.	Assignment 1 (UI) Assignment 2 (Data transfer)
Identification	A page that allows the user to upload a photo to identify the scientific name, common name, care tips, and location of the plant. Include a “Search” feature that allows the user to search for plant information.	Assignment 1 (UI) Assignment 2 (Enhancement)
Contributor	A page to display the contributed plant information from the registered user.	Assignment 1 (UI) Assignment 2 (Enhancement)
Contribution page	A user interface for user to fill in to contribute the plant data to the website.	Assignment 1 (UI) Assignment 2 (Form validation)
Enquiry Form	A form where the user can fill in if they have any enquiry about the plants.	Assignment 1 (UI) Assignment 2 (Form validation, Data transfer)